

NEW OBSERVATIONS ON THE MECHANISMS OF DYNAMIC STRAIN AGING AND OF JERKY FLOW†

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Abstract—The rate sensitivity of the flow stress was investigated as a function of pre-strain for a commercial nickel alloy (INCONEL 600) and for polycrystalline Al-1% Mg. over a wide temperature range encompassing the regime of jerky flow. Anomalies were observed which suggest that the dynamic strain aging mechanism operates over a wider temperature range than that in which jerky flow is found. A detailed analysis shows that a negative rate sensitivity is associated with the strain hardening contribution to the flow stress from the beginning of deformation. The total rate sensitivity becomes negative only after this strain hardening contribution dominates. This suggests an interpretation of the 'critical strain' for the appearance of jerky flow, which does not rely on the production of vacancies and moreover casts doubt on any interpretation that associates dynamic strain aging with the interaction of mobile solutes with the moving dislocations. Instead, a dislocation arrest model is used in which solute atoms diffuse along dislocations at forest intersections.

Résumé—Nous avons étudié la sensibilité à la vitesse de la contrainte d'écoulement en fonction de la prédeformation dans un alliage commercial à base de nickel (INCONEL 600) et dans du Al-1% Mg polycristallin, pour une gamme étendue de températures comprenant le régime de l'écoulement saccadé. Les anomalies que nous avons observées nous donnent à penser que le mécanisme de vieillissement dynamique opère dans une gamme de température plus large que celle dans laquelle on observe l'écoulement saccadé. Une analyse détaillée montre qu'une sensibilité négative à la vitesse est associée à la contribution de l'écrouissage à la contrainte d'écoulement, dès le début de la déformation. La sensibilité totale à la vitesse ne devient négative que lorsque cette contribution domine. Ceci suggère une interprétation de la 'déformation critique' pour l'apparition de l'écoulement saccadé qui ne repose pas sur la production de lacunes et qui de plus jette un doute sur toute interprétation qui associe le vieillissement dynamique et l'interaction entre des atomes mobiles de soluté et des dislocations en déplacement. Nous utilisons au contraire un modèle d'arrêt des dislocations dans lequel les atomes du soluté diffusent le long des dislocations, aux intersections de la forêt.

Zusammenfassung—Die Geschwindigkeitsempfindlichkeit der Fließspannung wurde an einer kommerziellen Nickellegierung (INCONEL 600) und an einer polykristallinen Legierung Al-1% Mg über einen weiten Temperaturbereich einschließlich des Bereiches des ruckweisen Fließens untersucht. Es wurden Anomalien beobachtet, die nahelegen, daß der dynamische Rekalterungsmechanismus in einem größeren Temperaturbereich als demjenigen des ruckweisen Fließens abläuft. Eine genaue Auswertung zeigt, daß eine negative Geschwindigkeitsempfindlichkeit mit dem Verfestigungsbeitrag zur Fließspannung verbunden ist. Die gesamte Geschwindigkeitsempfindlichkeit wird nur negativ, wenn dieser Verfestigungsbeitrag überwiegt. Das legt eine Deutung der 'kritischen Dehnung' für das Auftreten ruckweisen Fließens nahe, die keinen Gebrauch macht von Leerstellenproduktion; überdies zieht diese jede Interpretation in Zweifel, die dynamische Rekalterung mit der Wechselwirkung beweglicher Fremdatome mit gleitenden Versetzungen verbinden. Stattdessen wird ein Modell der Versetzungshemmung benutzt, bei dem Legierungsatome an Kreuzungspunkten mit Waldversetzungen entlang der Versetzung diffundieren.

1. INTRODUCTION

The mechanism of jerky (or 'serrated') flow has long been an area of intensive research, and was the subject of several recent reviews [1-3]. Although it has been over 25 years since Cottrell proposed a model based on the immobilization of dislocations due to solute 'atmospheres' [4], there is as yet no clear consensus as to the details of the mechanism, although the Cottrell model is almost universally agreed to be involved in one form or another.

The present work is a study of the rate sensitivity of deformation in materials which exhibit jerky flow. The studies were carried out over a wide temperature range, although particular attention was given to temperatures and strain-rates just slightly below the regime in which jerky flow was found. At these temperatures a negative strain-rate sensitivity is not observed, the stress-strain curve is smooth, and the results are amenable to analysis by the conventional theory of thermal activation.

A chief question posed in thermal activation analysis is, how many mechanisms contribute significantly to the flow stress. A particularly easy case is when

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the flow stress has two additive contributions:

$$\sigma = \sigma_f + \sigma_d \quad (1)$$

For solution strengthened alloys, this should be applicable [5], with σ_f relating to a 'friction' imposed by the solutes on the mobile dislocations, and σ_d relating to the dislocation-dislocation interaction which is the cause of strain hardening.

The strain-rate sensitivity of the flow stress in such a material also has two additive contributions:

$$\left. \frac{\partial \sigma}{\partial \ln \dot{\epsilon}} \right|_T = \left. \frac{\partial \sigma_f}{\partial \ln \dot{\epsilon}} \right|_T + \left. \frac{\partial \sigma_d}{\partial \ln \dot{\epsilon}} \right|_T \quad (2)$$

The combination of equations (1) and (2) allows one to separate the two mechanisms by investigating the rate sensitivity as a function of strain, because only the second term in equation (2) should be influenced by work hardening.

In particular, if we write

$$\sigma_d = \alpha(\dot{\epsilon}, T) \cdot \mu b \sqrt{\rho}, \quad (3)$$

where ρ is the dislocation density that changes with strain and α describes the dependence of the effective interaction strength on thermal activation,

$$\left. \frac{\partial \sigma}{\partial \ln \dot{\epsilon}} \right|_T = \left. \frac{\partial \sigma_f}{\partial \ln \dot{\epsilon}} \right|_T + \frac{\sigma_d}{\alpha} \left. \frac{\partial \alpha}{\partial \ln \dot{\epsilon}} \right|_T \quad (4)$$

or

$$\left. \frac{\partial \sigma}{\partial \ln \dot{\epsilon}} \right|_T = \sigma_f \left(\left. \frac{\partial \ln \sigma_f}{\partial \ln \dot{\epsilon}} \right|_T - \left. \frac{\partial \ln \alpha}{\partial \ln \dot{\epsilon}} \right|_T \right) + \sigma \left. \frac{\partial \ln \alpha}{\partial \ln \dot{\epsilon}} \right|_T \quad (5)$$

Thus, a plot of $(\partial \sigma / \partial \ln \dot{\epsilon})_T$ vs flow stress (as originally proposed by Haasen [6]) will give a straight line; its slope is:

$$\frac{1}{m_d} \equiv \left. \frac{\partial \ln \sigma_d}{\partial \ln \dot{\epsilon}} \right|_T = \left. \frac{\partial \ln \alpha}{\partial \ln \dot{\epsilon}} \right|_T \quad (6)$$

and its intercept is given by the first term on the right of equation (5).

Equation (5) constitutes a generalization of one more commonly used [5, 7] which identifies the term $(\partial \sigma / \partial \ln \dot{\epsilon})_T$ with inverse activation areas. The study described in the following showed that the straight-line relation to equation (5) is obeyed over a significant range of variables and the results are analyzed with an intent to determine the existence and nature of two contributions to the flow stress. An unexpected feature of these results makes interpretation in terms of 'activation areas' inappropriate, as will be seen in the discussion.

2. EXPERIMENTAL PROCEDURE

A commercial Inconel 600 alloy (composition of 15.9 at.% Cr-10.1 at.% Fe-0.4 at.% C-bal. Ni) was obtained in the form of 12.7 mm dia. bar and machined into cubical compression samples, ~6 mm

on a side, by electrical discharge machining, with the compression axis aligned with the drawing direction. The compression faces of the samples were parallel within ± 0.008 mm. These samples were recrystallized at 1473 K and then given a programmed heat treatment designed to precipitate excess carbon. This was done to avoid complications arising from precipitation of carbides on dislocations during deformation, which has been observed in similar alloys [8]. The treatment consisted of a furnace cool from 1473 K to 1210 K, a slow cool to 1100 K over a 20 h period, a hold at 1100 K for 50 h and a furnace cool to room temperature. The resulting average grain size was 0.05 mm. All treatments were done using an inert high-purity Ar atmosphere and a Ta getter near the sample.

A high purity Al-1 at.% Mg alloy, vacuum cast from 99.99% purity Al and Mg and hot extruded was also obtained. These were machined as above into cubical compression samples ~8 mm on a side, annealed at 773 K and furnace cooled. The average grain size was 0.6 mm.

Compression tests were performed in a floor model Instron mechanical testing machine at strain-rates ($\dot{\epsilon}$) from $2 \times 10^{-5} \text{ s}^{-1}$ to $2 \times 10^{-3} \text{ s}^{-1}$. All stresses are reported as true axial stresses, all strains as true axial strains. The load train was designed to maintain a high machine stiffness ($\sim 50 \text{ kN mm}^{-1}$). The loading rams were rigid and the loading faces were aligned parallel within 0.001 rads. At low temperatures ($200 \text{ K} \leq T \leq 300 \text{ K}$) samples were loaded between hardened steel platens lubricated with molybdenum disulfide powder in a specially constructed cryostat which used a constant flow of liquid nitrogen to cool the pushrods and chamber and a temperature-controlled heater which balanced the constant cooling rate and maintained the desired temperature ($\pm 0.5 \text{ K}$). At elevated temperatures ($350 \text{ K} \leq T \leq 1000 \text{ K}$) a three-zone resistance furnace was used. Tests at temperatures between 350 K and 500 K were done in air with a loosely fitting quartz tube surrounding the pushrods and sample to reduce convection currents and improve temperature stability. Above 500 K tests were done in a flowing N_2 atmosphere, with polished Al_2O_3 or Si_3N_4 discs as load platens and 0.1 mm thick Au foil as a lubricant. Tests with and without the Au foil showed no detectable deformation or relaxation due to the foil and its use significantly reduced barreling.

The strain-rate sensitivity was measured using strain-rate change tests. Strain-rate changes with $\dot{\epsilon}_1/\dot{\epsilon}_2 = 10$ or 0.1 were effected using the Instron variable speed control. No difference was observed in the results from upward or downward changes in the strain-rate. A separate x-y recorder was used to measure load and elongation with high resolution during these tests. The extrapolation used to calculate strain-rate sensitivities is defined in Fig. 1. As also pointed out by van den Brink *et al.* [9], $\Delta\sigma$ (rather than $\Delta\sigma_i$) is the appropriate measure of the strain-rate

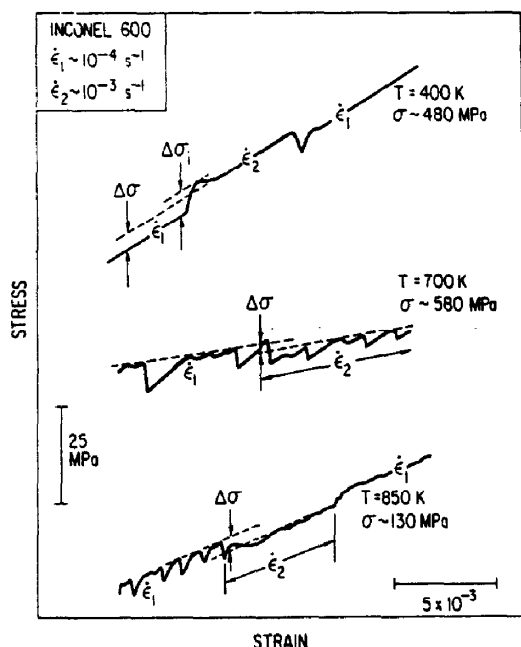


Fig. 1. Illustration of the results of strain-rate change tests at various temperatures for Inconel 600. The $\Delta\sigma$ extrapolation shown was used for the calculation of strain-rate sensitivity. When jerky flow was present the extrapolations were drawn tangent to the stress peaks. The nominal stress level is also indicated for each curve.

sensitivity of the flow stress under dynamic strain aging conditions.

3. RESULTS

Jerky flow was exhibited by the Inconel 600 samples at temperatures above about 500 K. Some

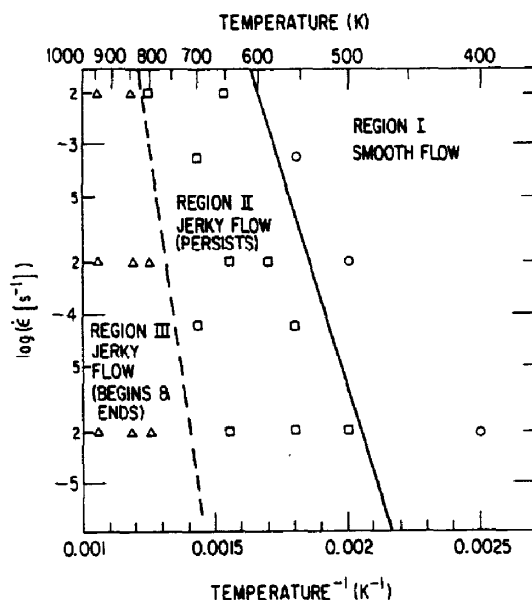


Fig. 2. The strain-rate and temperature regime of jerky flow in Inconel 600.

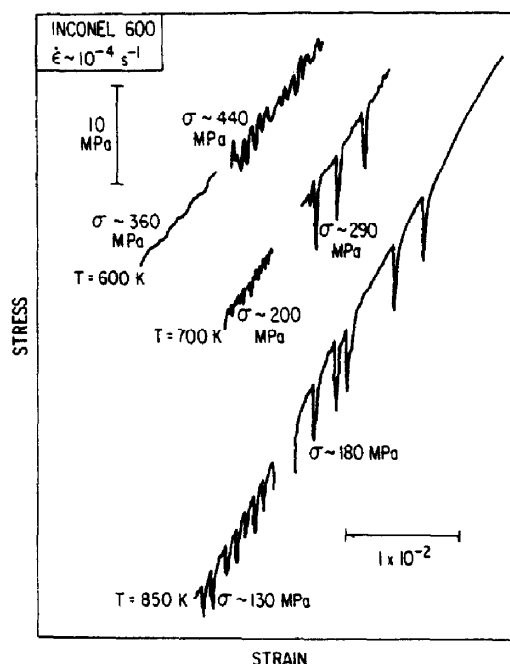


Fig. 3. Example of the different types of jerky flow observed in Inconel 600 tested in compression, including its disappearance at large strain at 850 K. The temperature and nominal stress level of each curve is indicated.

jerky flow was found even at the highest test temperature, 1000 K, although its extent was very limited, consisting of only a few serrations at the beginning of the stress-strain curve. Figure 2 summarizes the strain-rate and temperature regime in which jerky flow was observed. Three regions may be identified. In region I (high $\dot{\epsilon}$ or low T), jerky flow was never observed. In region II, jerky flow began after a specific amount of strain had been accumulated (generally a few per cent, depending on the strain-rate and temperature) and was present throughout the test which continued until the stress-strain curve was near saturation. In region III, which existed at the highest test temperatures, jerky flow would commence shortly after yield but continue only for a brief amount of strain. The temperature regime of jerky flow in Inconel 600 is higher than that of binary Ni-C alloys [10] and is similar to the results of Jenkins and Smith on austenitic alloys in which it was concluded that diffusion of a Cr-[C, N] complex was responsible [11]. (Jerky flow is not generally observed in Ni alloys in the absence of interstitials such as C or N [12].)

Typical examples of jerky flow in Inconel 600 are shown in Fig. 3. The stress-strain curve was sometimes characterized by successive, regularly spaced load drops of a saw-tooth nature, especially at low temperatures or strains (e.g. 600 K at a stress of 440 MPa). At high temperatures or strains a relatively smooth flow curve, punctuated by sudden load drops, was often observed (e.g. 850 K at a stress of 180 MPa).

These serrations are similar to the 'fine' or type

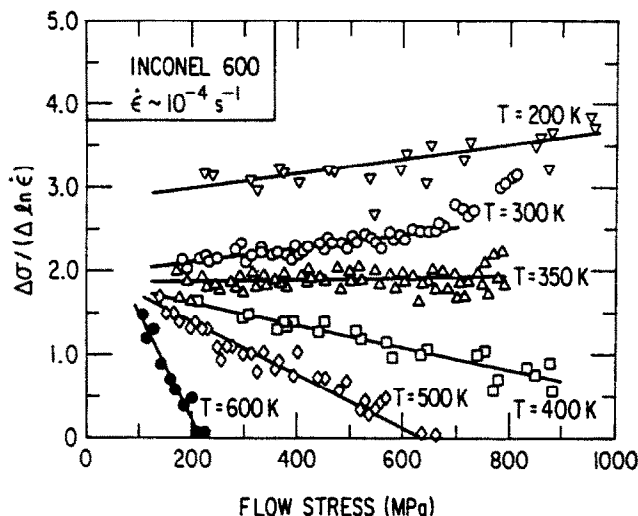


Fig. 4. Behavior of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs flow stress at low and moderate temperatures for Inconel 600.

II serrations observed in other systems [13, 14]. Serrations of 'periodic' or type I nature [13], in which there is a sudden increase in work hardening rate and a stress peak above the average level of the stress-strain curve followed by a discontinuous drop in stress and then a portion of smooth deformation, were never observed.

The results of a study of the strain-rate sensitivity of Inconel 600 are shown in Fig. 4. At temperatures of 300 K or lower the behavior is 'normal': $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ increases with increasing flow stress. At 400 K, however, the behavior is 'anomalous': $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ decreases linearly with increasing flow stress. The observed strain-rate sensitivity remains positive, however, and the stress-strain curve is smooth. At somewhat higher temperatures (e.g. 500 K), $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ decreases to zero and if deformation is continued a negative strain-rate sensitivity is observed. Jerky flow begins at the point where the rate sensitivity becomes negative or very shortly thereafter; jerky flow was never observed in a regime where the strain-rate sensitivity was positive, although smooth flow was sometimes observed when the strain-rate sensitivity had a small negative value. It has been noted previously that negative strain-rate sensitivity is a characteristic feature of materials deformed under conditions producing jerky flow [15-17].

If deformation is continued into the region of negative strain-rate sensitivity, the $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs flow stress plots are no longer linear, as shown in Fig. 5. At temperatures of about 850 K and above (corresponding to temperatures and strain-rates which lie in region III of Fig. 2) a modified behavior was observed, consisting of a rapid decrease of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ through zero, followed by a subsequent increase to positive values again. This behavior (at 960 K) is also illustrated in Fig. 5. Jerky flow was generally observed during the portions of the stress-strain curves where the rate sensitivity was negative, although as noted

previously smooth flow was sometimes observed when $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ was negative but small. For example, at 960 K jerky flow was observed only near the minimum of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$.

The yield stress σ_y of annealed Inconel 600, and the saturation stress in compression σ_s vs temperature are shown in Fig. 6, normalized by the shear modulus μ . The saturation stress was defined by extrapolating the plot of the work hardening rate $\theta \equiv d\sigma/d\epsilon$ vs σ to zero stress and the shear modulus of Inconel 600 was approximated by that of Ni [18]. Note that σ_s/μ shows a peak within the temperature range where jerky flow occurs, while the yield stress shows only a broad plateau. A similar behavior has been observed for the UTS of Inconel 600 [19]. The work hardening data for various temperatures are shown in Fig. 7. The shape of the θ vs σ relation differs significantly at temperatures above and below the hump in the σ_s vs T curve.

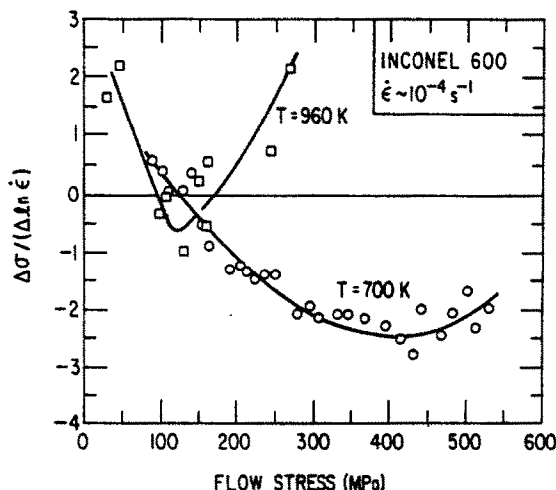


Fig. 5. Behavior of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs flow stress at high temperatures for Inconel 600.

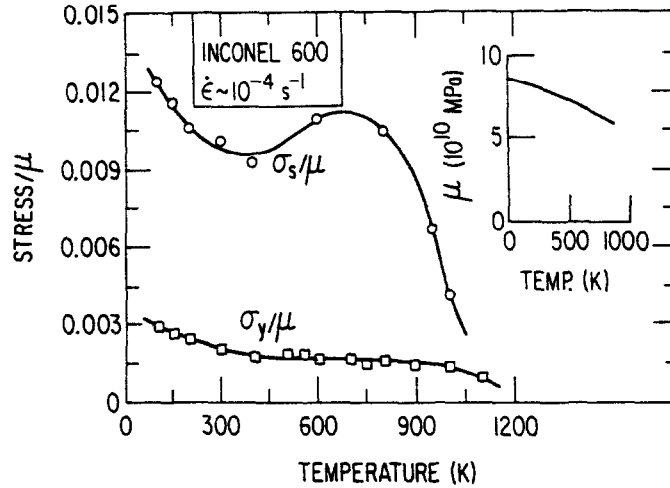


Fig. 6. Temperature dependence of the yield stress σ_y and the saturation stress σ_s , normalized by the shear modulus, for Inconel 600. The temperature dependence of the shear modulus of Inconel 600 was approximated using data for Ni [18].

A similar behavior of the strain-rate sensitivity was found in Al-1% Mg. A plot of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs flow stress at low and moderate temperatures is shown in Fig. 8. Irregular stress-strain curves were observed during compression testing in the region where the strain-rate sensitivity was negative. While regularly spaced 'fine' serrations were sometimes found, a more common form of the stress-strain curve in the region of negative rate sensitivity consisted of a series of plateaus rather than actual serrations (see Fig. 9). Well developed serrations have been observed during tensile testing of a similar alloy in this temperature range [20] and it is possible that the compression test suppresses the load drops.

At temperatures above 300 K, a different behavior was observed in the $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ plots: $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ is near zero during the initial portion of the stress-strain

curve, after which it increases rapidly with further increase in stress (Fig. 10). This temperature range corresponds to the regime where a change in the type of serrations from Type I 'periodic' to Type II 'fine' has been observed in tensile samples of Al alloys [21, 22].

The flow stress (or alternatively the strain) necessary to initiate jerky flow depends not only on the test temperature but also on the strain-rate. If the initiation of jerky flow is correlated with a transition from positive to negative rate sensitivity, it follows that the measured strain-rate sensitivity must itself be a function of strain-rate. Plots of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs flow stress, in which strain-rate changes were made from different base strain-rates are shown for Inconel 600 in Fig. 11. In this figure (for a test temperature within region I of Fig. 2) the slope of the $\Delta\sigma/(\Delta \ln \dot{\epsilon})$

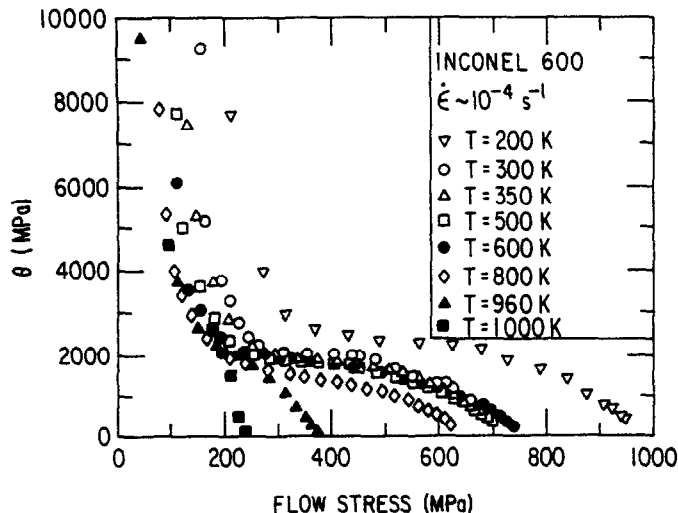


Fig. 7. Work hardening rate θ vs flow stress at various temperatures for Inconel 600.

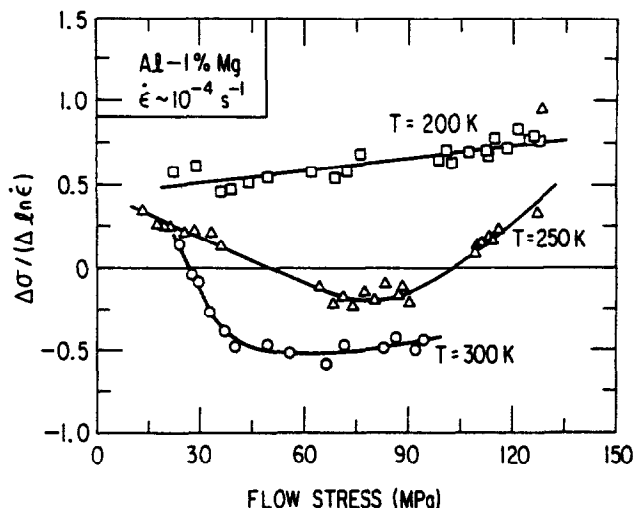


Fig. 8. Behavior of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs flow stress at low and moderate temperatures for an Al-1 at.% Mg alloy.

plot is negative at the lower strain-rate but positive at the higher. At higher temperatures similar effects of strain-rate on $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ were found. For example, at 850 K (a temperature within region III of Fig. 2) jerky flow continued up to higher flow stresses at a strain-rate of 10^{-4} s^{-1} than at 10^{-5} s^{-1} . Consequently at certain stress $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ was positive if measured from a base strain-rate of 10^{-5} s^{-1} but negative if measured from a base strain-rate of 10^{-4} s^{-1} . A similar phenomenon was observed in Al-1% Mg, as illustrated in Fig. 12.

4. DISCUSSION

A. The Cottrell model of jerky flow

It is widely believed that jerky flow arises from the interaction of diffusing solute atoms with mobile dislocations, as first proposed by Cottrell [4]. Since the diffusion rates of most substitutional solutes at temperatures where jerky flow occurs are too small to account for the effect, Cottrell proposed that the pre-strain required to initiate jerky flow (the 'critical strain' ϵ_c) was necessary in order to increase the vacancy concentration and thus enhance diffusion. Measurements of the 'critical strain' have figured prominently in investigations of jerky flow since this time. A correct interpretation of the critical strain is crucial to the interpretation of these works and a number of elaborations have been made on the original Cottrell model in order to better explain the data. Nevertheless it has not been possible to satisfactorily account for all the experimental observations, as pointed out recently by Baird [3]. For example, if a material is prestrained to a value of half that required to initiate jerky flow and then rested at the test temperature or annealed at a temperature high enough to annihilate excess vacancies but low enough that the dislocation structure is not affected, the total

pre-strain necessary to initiate jerky flow should be increased. This does not seem to occur [23, 24]. It should be noted here that experiments in which a metal is quenched from a high annealing temperature generally do show a reduced ϵ_c compared with slowly cooled samples [25, 26], or at least an effect on the types of serrations that occur [27]. However, the interpretation of these experiments is complicated by the possible occurrence of effects such as divacancy formation, vacancy clustering, the annihilation of vacancies at dislocations resulting in jogs, etc. Another problem is that values derived for the activation energy for jerky flow using the Cottrell model show a dependence on grain size [22], unless some rather specific assumptions are made about the relationship between grain size, strain and dislocation

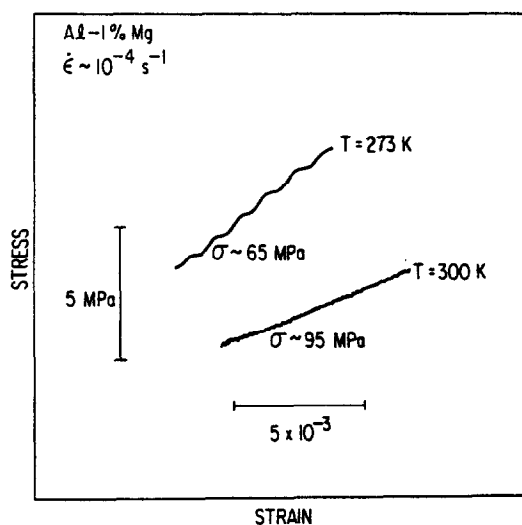


Fig. 9. Examples of jerky flow observed in Al-1 at.% Mg tested in compression. The temperature and nominal stress level of each curve are indicated.

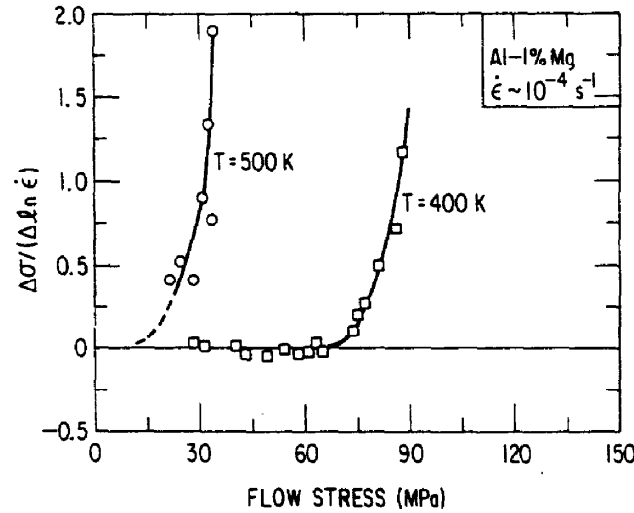


Fig. 10. Behavior of $\Delta\sigma/(\Delta\ln\dot{\epsilon})$ vs flow stress at high temperature for Al-1 at.% Mg.

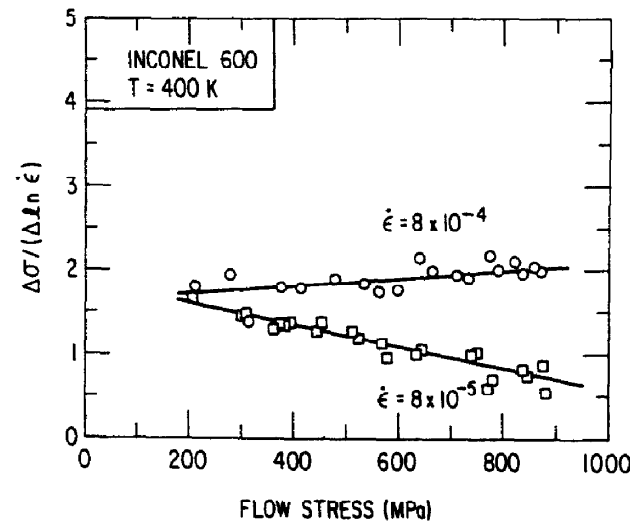


Fig. 11. Effect of strain-rate on the $\Delta\sigma/(\Delta\ln\dot{\epsilon})$ vs flow stress relationship at low temperatures in Inconel 600.

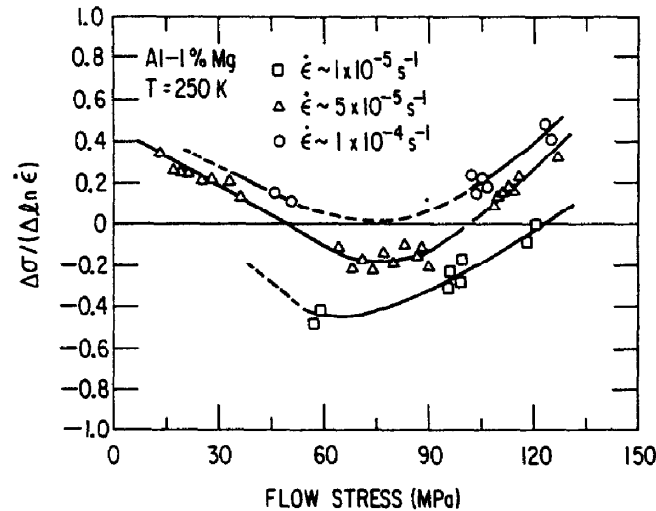


Fig. 12. Effect of strain-rate on the $\Delta\sigma/(\Delta\ln\dot{\epsilon})$ vs flow stress relationship in Al-1 at.% Mg.

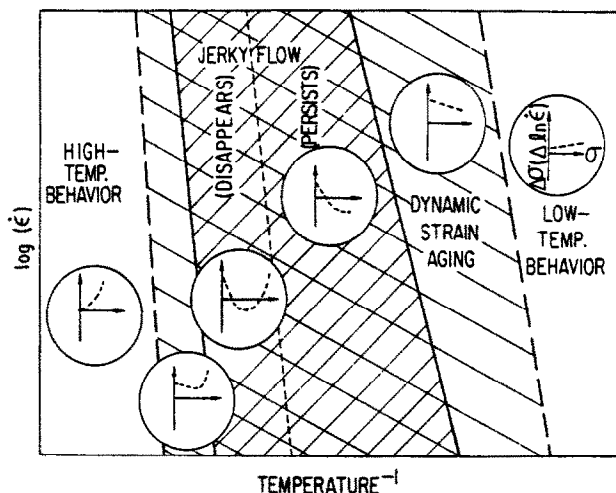


Fig. 13. Schematic diagram of the temperature and strain-rate regimes of dynamic strain aging and of jerky flow. The behavior of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs σ in each region is indicated.

density [28]. Suffice it to say that the evidence supporting the current interpretation of the role of ϵ_c is ambiguous at best.

B. Interpretation of the $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs flow stress plots

In the plots of $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs σ , an anomalous negative slope is observed in a temperature regime which not only includes the region in which jerky flow is observed but extends to somewhat lower temperatures as well. Also, this anomalous negative slope exists from the beginning of deformation (where the rate sensitivity is positive) although jerky flow does not occur until after some deformation is accumulated and the rate sensitivity becomes negative. A similar result for an Al-0.7% Mg alloy was reported by MacEwen and Ramaswami [29]. The anomalous negative slope implies that

$$1/m_d \equiv \left. \frac{\partial \ln \alpha}{\partial \ln \dot{\epsilon}} \right|_T,$$

is negative in equation 5 and this cannot be understood in terms of any of the usual low-temperature strain hardening mechanisms; a new mechanism is required.

An attractive interpretation is that the negative slope is due to the operation of the dynamic strain aging mechanism. If this is the case, it is likely that the mechanism involves solute interaction with the strain hardening (since it appears to be m_d rather than the rate sensitivity of the friction stress which is affected). Since the mechanism must be operating from the beginning of deformation, vacancies produced during deformation would not play an important role. A model which has these features will be discussed in the next section. Although the proposed mechanism cannot by itself produce jerky flow, its contribution to the total rate sensitivity is negative and increases with strain. Thus, although the total rate sensitivity is initially positive (due to the contri-

bution of the σ_f term in equation 5), it will eventually become negative when sufficient strain has been accumulated to make σ/m_d the dominant term in equation (5).

Experimentally, it is well established that a negative rate sensitivity is correlated with the occurrence of jerky flow [9, 15-17]. In fact, the appearance of jerky flow can be explained on this basis alone, without reference to the underlying physical mechanisms [30]. This provides a convenient interpretation of the critical strain: it is the strain necessary to produce a negative rate sensitivity (or a critical negative value of the rate sensitivity as suggested by van den Brink *et al.* [9]).

The above interpretation of dynamic strain aging might also explain the frequent observation that dynamic strain aging affects the work hardening rate [3] and that it has a stronger effect on the saturation stress (or the UTS) than on the yield stress [19], since it is the strain hardening process which is affected by the dynamic strain aging.

There have been other observations which suggest that the operation of the dynamic strain aging mechanism does not necessarily entail jerky flow. For example, Brindley and Worthington [22] found that jerky flow occurred at higher temperatures in single crystals of Al-3% Mg than in polycrystals despite the fact that the high temperature drop in the flow stress occurred at the same temperature in the polycrystals as in the single crystals, indicating that the same dislocation mechanisms were operative in both types of samples.

The conclusion which is reached in this section may be summarized by modifying Fig. 2 so that it displays the region in which the dynamic strain aging mechanism occurs. The region in which jerky flow is found is distinct but is contained within this region. This is shown schematically in Fig. 13. Decerf and Bauwens have published a phenomenological descrip-

tion of serrated flow based on the mathematical formalism of non-Newtonian viscosity which yields a similar result [31].

C. A model for dynamic strain aging

The required features of the mechanism of dynamic strain aging are provided by a model based on the dislocation arrest theory originally proposed by Sleswyk [32]. This theory suggests that solute atmospheres form on forest dislocations and then drain by pipe diffusion from the forest dislocations to the mobile dislocations during the time spent by the mobile dislocations while they are waiting at the forest dislocations. Since this model does not depend on bulk diffusion to form the solute atmospheres on dislocations, the apparent conflict between diffusion rates and the temperature and $\dot{\epsilon}$ regime of jerky flow is resolved. Dislocation pipe diffusion should be rapid enough to allow the formation of atmospheres on dislocations in this temperature range without requiring any enhancement, especially since the solute atmosphere need not pin the entire dislocation segment to have an effect, but merely a portion of the dislocation line at the forest dislocation junction. Thus very little diffusion need be involved to produce obstacles which have a strength that increases with waiting time, resulting in a negative m_d . Dislocation pipe diffusion was suggested as a factor in the process causing serrations in Ni-C alloys by Nakada and Keh [10] and in (ferritic) substitutional solid solutions of Fe by Cuddy and Leslie [33].

An important feature of the present interpretation is based on the fact that it is m_d rather than the friction stress which is affected by the dynamic strain aging. This implies that it is the strength of the dislocation-dislocation interactions which is altered by the mechanism. One way this could occur is through the effect of the solute atmospheres on the stacking fault energy (Suzuki locking). In support of this, it has been shown that the stacking fault energy does affect m_d in Ni-Co alloys and in Inconel 600 [7]. Ordinary 'elastic' locking could also be responsible if solute diffusing along the 'mobile' dislocation away from the junction has the effect of making the forest dislocation a 'bigger' obstacle.

It should be noted that McCormick [34] has proposed a modified Sleswyk model which applies to f.c.c. substitutional alloys. This model has been developed in detail by van den Beukel *et al.* [35, 9]. Experiments in which the strain-rate sensitivity of Au-Cu alloys was investigated [35] have led these workers to conclude, in a manner similar to that of the present study, that jerky flow initiates only after the dynamic strain aging mechanism becomes sufficiently effective (through the accumulation of strain) that a negative strain-rate sensitivity is produced. However, unlike the present model, they assume that atmospheres are formed on (mobile) dislocations by bulk diffusion during the waiting time, and that the strain dependence of jerky flow arises from an enhancement of the diffu-

sion coefficient by vacancy production during deformation. This interpretation is open to the same objections to the importance of strain enhanced diffusion as were leveled against the Cottrell model in a previous section. Moreover, the straight-line character of the $\Delta\sigma/(\Delta \ln \dot{\epsilon})$ vs σ plots would be rather fortuitous in this model. It is believed that a more consistent picture is provided by the present interpretation.

One consequence of the proposed mechanism of dynamic strain aging merits further elaboration. It is generally found that the activation energies calculated for the initiation and for the disappearance of jerky flow differ [36]. If the present model is correct these activation energies have questionable physical significance and there is no compelling reason that either should be the same as the activation energy for dynamic strain aging. Studies of the *mechanism* of jerky flow, however, must be concerned with the latter quantity. In particular, procedures in which an activation energy is derived from the temperature and strain-rate dependence of the critical strain need to be re-evaluated.

5. CONCLUSION

The strain-rate sensitivity of the flow stress shows anomalies in a range of temperatures not normally associated with dynamic strain aging and also at strains smaller than the 'critical strain' for the appearance of jerky flow. These observations suggest that the dynamic strain aging *mechanism* is present over a wider range of the variables, but becomes gradually more important as the condition of macroscopic jerky flow is approached.

A particular way in which dynamic strain aging could become more effective as straining proceeds is through an influence on the strain hardening contribution to the flow stress, rather than on the friction stress. The experiments suggest just such an interpretation: the rate sensitivity of strain hardening is negative from the beginning; after some ('critical') strain it dominates and renders the total rate sensitivity negative; and this in turn causes jerky flow.

A modified Sleswyk model could explain the influence of solute mobility on strain hardening: the mobile dislocations drain solutes from the forest dislocations while they are waiting for thermal activation and the cutting mechanism itself is thereby strengthened. Such a model could also explain both the 'critical strain' for jerky flow and the higher strain hardening coefficients usually associated with alloys that show dynamic strain aging.

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